

Phosphorus Partition and Phosphate Capacity of Basic Oxygen Steelmaking Slags

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The phosphorus partition (L_P) and phosphate capacity ($C_{PO_4^{3-}}$) were measured for slags in the CaO–SiO₂–MgO–Fe₂O₃–(MnO–Al₂O₃–TiO₂–P₂O₅) system over temperatures representative of basic oxygen steelmaking. The measured L_P values were found to be in good agreement with those predicted from the model of Assis *et al.* The oxygen potential of the slag-metal systems studied was also evaluated and used in combination with the measured L_P to calculate the $C_{PO_4^{3-}}$ of the slags. Capacity values in the range of 7.77×10^{16} to 4.27×10^{19} for temperatures 1 550°C to 1 700°C were obtained.

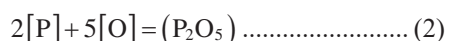
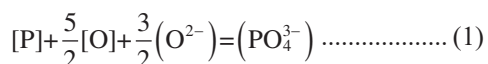
Correlations of $C_{PO_4^{3-}}$ with common measures of slag basicity (v-ratio and optical basicity) were sought and reported. Consistent with other researchers it was found that the L_P and $C_{PO_4^{3-}}$ increased with increasing slag basicity and decreasing temperature.

KEY WORDS: basic oxygen steelmaking (BOS); dephosphorisation; phosphorous partition; phosphate capacity; laboratory data; slags.

1. Introduction

Phosphorus generally has a negative effect on the mechanical properties of steel.^{1–3)} Phosphorous levels in iron ore are expected to increase^{4,5)} and may cause difficulties in achieving steel specification using current processing methods and conditions. Against this background of increasing phosphorous in the ore, an improved understanding of steel dephosphorisation fundamentals will underpin any adjustments steelmakers need to make to the steelmaking process to continue to meet steel specification.

Phosphorus removal in basic oxygen steelmaking (BOS) is typically described by the ionic reaction given in Eq. (1). It may also be represented by the simple molecular reaction, given in Eq. (2),



where () denotes solution in slag and [] denotes solution in liquid iron. The reference state for the metal solutes is the 1 mass% standard state and for (P₂O₅) is the hypothetical molten pure state.

The Gibbs free energy relation for the reaction given in

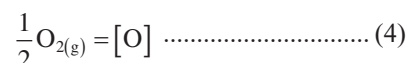
Eq. (2) is given in Eq. (3)⁶⁾

$$\Delta G^\circ = -832\,384 + 632.65T \quad \text{J} \cdot \text{mol}^{-1} \quad \text{..... (3)}$$

where T is the temperature in K.

It can be seen from the reactions given Eqs. (1) and (2), as well as Eq. (3), that phosphorus removal is promoted by high [P] activity, high oxygen potential (represented as [O] in Eqs. (1) and (2)), high basicity (O²⁻) and lower temperature (T). Unfortunately, for integrated steelmakers, there is no location in the iron and steelmaking process where all of these conditions are met. Dephosphorisation is typically carried out in either hot metal pre-treatment that has process characteristics of low T, high O²⁻ but low [O] or in the BOS converter that has process characteristics of high T, high O²⁻ and high [O]. It can also be removed in secondary refining units where the basicity and [O] can be tailored to suit steel specification requirements.

While the reactions given in Eqs. (1) and (2) consider the oxygen source for oxygen potential in the form of [O], it can be represented by whatever reaction, or balance of reactions, which control the delivery of oxygen for the phosphorous reaction. [O] can be related to the partial pressure of oxygen (pO₂) by consideration of the reaction given in Eq. (4) and its respective Gibbs free energy in Eq. (5).⁶⁾



$$\Delta G^\circ = -115\,750 - 4.63T \quad \text{J} \cdot \text{mol}^{-1} \quad \text{..... (5)}$$

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While the basicity of a slag infers a structure (or degree of polymerisation), there is no universal approach to basicity that entirely represents structure through a broad compositional range from acid to base that is fully quantitative. This has resulted in a number of empirical and semi-empirical approaches to represent basicity. In the dephosphorisation literature the v -ratio and optical basicity (Λ), defined in Eqs. (6) and (7),^{7,8} are often used as convenient representations for assessing the effect of basicity on dephosphorisation⁹

$$v = \frac{(\% \text{CaO})}{(\% \text{SiO}_2)} \dots\dots\dots (6)$$

$$\Lambda = \sum_i^j X_i \Lambda_i \dots\dots\dots (7)$$

where % i is the mass% of the oxide i , Λ_i is the optical basicity of oxide i and X_i is the equivalent cation fraction of oxide i .

The v -ratio is the most commonly used basicity approximation used in industry. Single oxide values, Λ_i , recommended in a review by Sommerville *et al.*¹⁰ and used in Eq. (7) are given in **Table 1**.¹⁰

The phosphorus partition (L_P) given in Eq. (8), is commonly used to assess the dephosphorisation ability of slags.

$$L_P = \frac{(\% \text{P})}{[\% \text{P}]} \dots\dots\dots (8)$$

Although the L_P is a useful approach its use is limited to conditions with comparable oxygen potential ranges under which the L_P was measured.¹¹ In the many published L_P models that predict the partition from slag compositions, the oxygen potential is often accounted for by some proxy, such as Fe_t or Fe_tO .^{9,12}

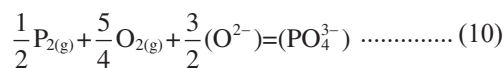
A recently published L_P model by Assis *et al.*¹³ is given in Eq. (9)

$$L_P = 0.068[(\% \text{CaO}) + 0.42(\% \text{MgO}) + 1.16(\% \text{P}_2\text{O}_5) + 0.2(\% \text{MnO})] + \frac{11570}{T} - 10.52 + 2.5 \log(\% \text{Fe}_t) \dots\dots (9)$$

where T is the temperature in K and $\% \text{Fe}_t$ is the total iron oxide in slag represented as elemental Fe in mass percent.

This model represents a reviewed and updated version of the Suito *et al.*¹⁴⁻¹⁶ model that has incorporated the data of Basu *et al.*^{17,18} and Assis *et al.*^{13,19} It was developed using laboratory data representative of industrial BOS slags.

A slag's dephosphorising potential can also be quantified by considering its phosphate capacity. The gas-slag reaction in Eq. (10) can be used to define the phosphate capacity ($C_{\text{PO}_4^{3-}}$) of a slag, as shown in Eq. (11),²⁰



$$C_{\text{PO}_4^{3-}} = \frac{(\% \text{PO}_4^{3-})}{p_{\text{P}_2}^{1/2} p_{\text{O}_2}^{5/4}} = \frac{K_{\text{PO}_4^{3-}} a_{\text{O}^{2-}}^{3/2}}{f_{\text{PO}_4^{3-}}} \dots\dots\dots (11)$$

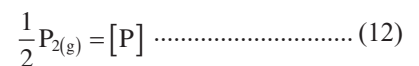
Table 1. Optical basicity of oxides recommended by Sommerville *et al.*¹⁰

Oxide	CaO	MnO	FeO	MgO	Fe ₂ O ₃	Al ₂ O ₃	TiO ₂	SiO ₂	P ₂ O ₅
Λ_i	1.00	0.95	0.93	0.85	0.69	0.65	0.65	0.48	0.4

where $(\% \text{PO}_4^{3-})$ is the mass% of phosphate ions in the slag, $K_{\text{PO}_4^{3-}}$ is the equilibrium constant for PO_4^{3-} formation of reaction given in Eq. (10), p_i is the partial pressure of species i , f_i is the activity coefficient of species i and $a_{\text{O}^{2-}}$ is the activity of O^{2-} in the slag.

In the $C_{\text{PO}_4^{3-}}$ calculation it is assumed that the activity coefficient for phosphate ion remains constant in the slag with increasing phosphorus content and that the oxygen ion activity is independent of the oxygen potential. The $C_{\text{PO}_4^{3-}}$ of various slags reported in the literature have been reviewed in Thermodynamic Data for Steelmaking²¹ and the Slag Atlas.²²

The $C_{\text{PO}_4^{3-}}$ defined in Eq. (11) can be derived from the L_P by considering the reaction given in Eqs. (12), and (13),⁶



$$\Delta G^\circ = -122173 + 19.25T \quad \text{J} \cdot \text{mol}^{-1} \dots\dots\dots (13)$$

and is defined in Eq. (14)

$$C_{\text{PO}_4^{3-}} = \frac{L_P K_{[\text{P}]} M_{\text{PO}_4^{3-}}}{p_{\text{O}_2}^{5/4} f_{\text{P}} M_{\text{P}}} \dots\dots\dots (14)$$

where M_i is the molecular weight of species i and $K_{[\text{P}]}$ is the equilibrium constant for the reaction given in Eq. (12).

The objective of this study is to examine the dephosphorisation ability of typical BOS steelmaking slags in the $\text{CaO-SiO}_2\text{-MgO-Fe}_t\text{O}$ system at temperatures representative of industrial practice. This study expands on the limited $C_{\text{PO}_4^{3-}}$ and L_P data available (~24 data points) in the literature^{15,16,18,23} at industrial temperatures (~1 650°C) and slag basicities (v -ratio 2.5–5).

2. Experimental

A series of L_P experiments were carried out in a vertical tube furnace under a controlled Ar atmosphere for 12 hours. A schematic of the set-up used is shown in **Fig. 1**. The L_P experiments used 4 g of slag and 30 g of 0.18 mass% [P] alloy melted in a high purity dense MgO crucible (19.3 mm ID, 31.75 mm high). During heating, a Fe foil cup

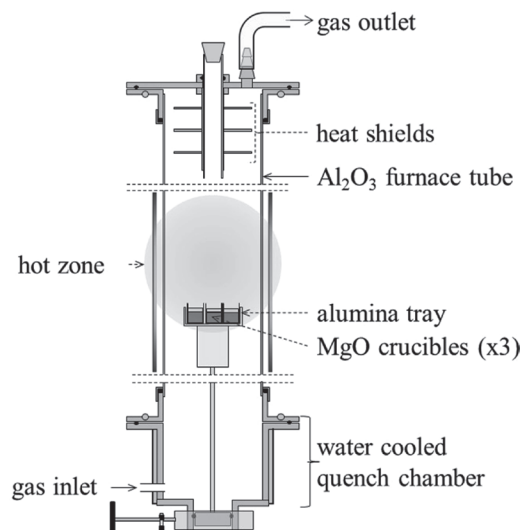


Fig. 1. A schematic of the experimental furnace set up used.

(~0.16 g) was used to keep the slag and alloy separate until both the slag and alloy were fully liquid. The experimental temperature ranged from 1 550°C to 1 700°C. The slag compositions and temperatures used were chosen to be representative of industrial BOS practice. The high purity (99.999%) Ar gas was scrubbed prior to entering the furnace by passing it through ascarite, drierite and then copper turnings at 300°C.

Once cooled to room temperature the crucibles were removed from the furnace and the slag and metal was mechanically separated in preparation for composition analysis. The metal was cleaned of any remaining slag and digested with nitric acid for ICP-OES analysis and analysed using an Agilent 710 series ICP-OES. The slag was crushed with a mortar and pestle for x-ray fluorescence (XRF) analysis using an AMETEK SPECTRO XEPOS energy dispersive polarisation XRF spectrometer. A selection of the crushed slag samples were also analysed for $\text{Fe}^{2+}/\text{Fe}^{3+}$ using an acid digestion and titration method. The digested slag was titrated against potassium dichromate using a Metrohm 798 MPT Titrino with Pt/Ag-AgCl electrodes. The $\text{Fe}^{2+}/\text{Fe}^{3+}$ analysis was conducted at the CSIRO Clayton laboratories, Victoria, Australia.

2.1. Slag Preparation

Sintered synthetic slag pellets in the $\text{CaO-SiO}_2\text{-MgO-Fe}_2\text{O}_3$ system were prepared using high purity oxide powders

Table 2. Synthetic slag compositions as measured by XRF.

Description	MgO	Al_2O_3	SiO_2	P_2O_5	CaO	TiO_2	MnO	Fe_2O_3^*	v-ratio
Pre-fused slag	13.1	0.0	43.9	0.0	43.1	0.0	0.0	0.0	1.0
B2	9.1	0.0	23.1	0.0	45.8	0.0	0.0	22.1	2.0
B2	9.6	0.0	22.4	0.0	46.5	0.0	0.0	21.6	2.1
B2	9.7	0.0	21.4	0.0	44.5	0.0	0.0	24.4	2.1
B2 #	9.7	0.0	21.5	0.0	45.5	0.0	0.0	23.3	2.1
R	9.9	0.0	17.6	0.0	49.4	0.0	0.0	23.1	2.8
B4	9.4	0.0	16.7	0.0	63.0	0.0	0.0	10.9	3.8
B4	9.1	0.0	15.7	0.0	60.3	0.0	0.0	14.9	3.8
B4	6.8	0.0	17.4	0.0	66.8	0.0	0.0	8.9	3.8
B4	8.5	0.0	14.7	0.0	56.6	0.0	0.0	20.2	3.9
B4	9.8	0.0	13.6	0.0	52.8	0.0	0.0	23.8	3.9
B4	9.7	0.0	14.6	0.0	57.2	0.0	0.0	18.6	3.9
B4	9.7	0.0	13.3	0.0	53.4	0.0	0.0	23.6	4.0
B4	9.5	0.0	13.6	0.0	55.7	0.0	0.0	21.2	4.1
B4	7.7	0.0	15.0	0.0	63.9	0.0	0.0	13.4	4.3
B5	9.6	0.0	12.2	0.0	58.3	0.0	0.0	19.9	4.8
B5	9.6	0.0	12.8	0.0	61.3	0.0	0.0	16.3	4.8
B5	7.4	0.0	12.8	0.0	62.0	0.0	0.0	17.8	4.9
B5	9.7	0.0	12.5	0.0	60.9	0.0	0.0	17.0	4.9
B5	9.0	0.0	13.0	0.0	63.9	0.0	0.0	14.1	4.9
B5	10.0	0.0	11.1	0.0	54.5	0.0	0.0	24.4	4.9
B5	9.2	0.0	10.8	0.0	60.4	0.0	0.0	19.5	5.6

*All iron oxide reported as Fe_2O_3

(>99.9%) and pre-fused slag. The pre-fused slag was prepared by mixing dried (or calcined) oxide powders in the required amounts, followed by melting the mixed powder in a Pt crucible for ~20 minutes at 1 600°C using a muffle furnace. The liquid slag was then quenched by pouring on to a thick steel plate. The slag was remelted and again quenched then crushed to a fine powder using a ring grinder. The slag pellets were prepared by mixing dried (or calcined) powders and the pre-fused slag in the required ratios and pressing into 4 g pellets before sintering for 12 hours at 1 000°C in a muffle furnace. X-ray fluorescence (XRF) analysis of the unreacted slag pellets and the pre-fused slag are given in **Table 2**.

2.2. Alloy Preparation

A Fe-P master alloy was prepared for this study with a

Table 3. Alloy compositions in mass% using ARC-OES and [P] results from ICP-OES.

Description	C	P	Mn	Si	Al	Ti	V	S
Alloy 0.18 mass% [P]	0.033	0.181	0.217	0.010	0.038	0.009	0.000	0.010

Only trace levels (<0.003 mass%) of V were detected in the experimental alloys.

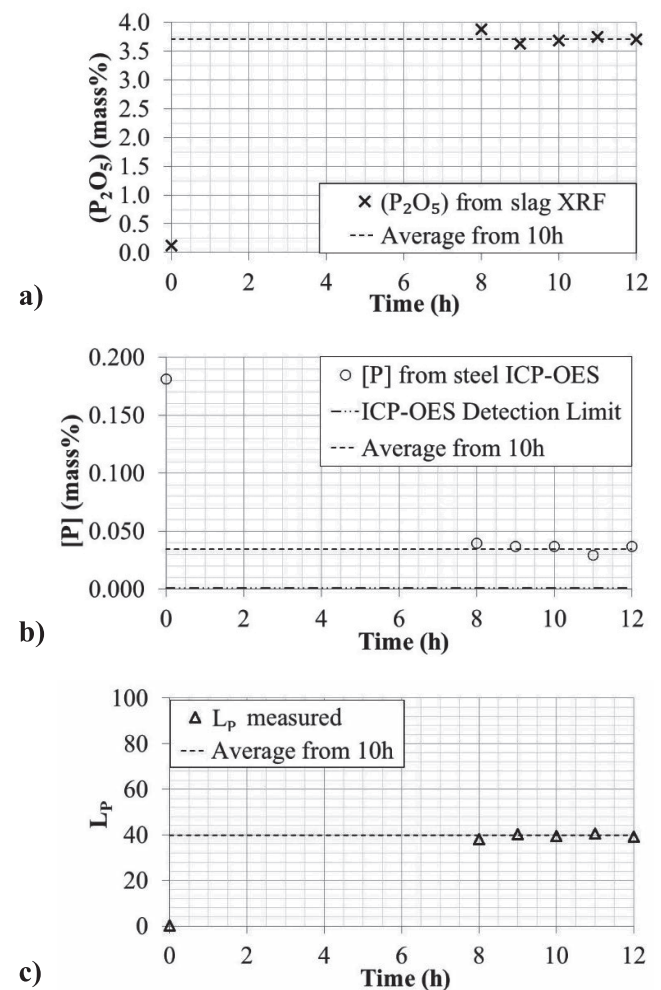


Fig. 2. Composition versus time plots for a) P_2O_5 measured via XRF, b) [P] measured via ICP-OES and c) L_p from measured P_2O_5 and [P].

[P] content of 0.18 mass%. The alloy was prepared using a ferro-phosphorus alloy (25.7 mass% [P]) and a low carbon, aluminium killed alloy (0.015 mass% [P]). The alloys were heated under an argon atmosphere to 1 600°C in an alumina crucible and held for 4 hours. The composition of the master alloy, determined via electric arc optical emission spectroscopy (ARC-OES) and inductively coupled plasma optical emission spectroscopy (ICP-OES), is given in **Table 3**.

2.3. Equilibrium Time

The equilibrium time for the L_P experiments was established by carrying out a series of experiments of up to 12 hours in duration under Ar using the B2 slag denoted by # in Table 2, with the Fe-P master alloy. The changes in the

phosphorous level were determined from both the metal and slag analysis and are given in **Fig. 2**. It was found that the measured (P_2O_5), [P] and L_P were constant, and at equilibrium, after 10 h reaction time. This is consistent with the 8 to 12 h equilibrium times reported in the literature.^{17,18,24)} To ensure equilibrium was achieved in this study, all experiments were run for at least 10 h, with the majority being run for 12 h.

3. Results and Discussion

3.1. L_P Evaluation

The slag and metal compositions measured in the L_P experiments are shown in **Table 4**. A series of seven experi-

Table 4. Experimental results for CaO–SiO₂–MgO–FeO–(MnO–Al₂O₃–TiO₂–P₂O₅) slags.

Exp.	Temp	Time	(MgO)	(Al ₂ O ₃)	(SiO ₂)	(P ₂ O ₅)	(CaO)	(TiO ₂)	(MnO)	(Fe _t O)*	Fe ²⁺	[P]	L _p	
No./Series	°C	h	mass%	mass%	mass%	mass%	mass%	mass%	mass%	mass%	mass%	mass%	ratio	
1	R	1 650	10	9.8	0.4	17.0	1.4	47.2	0.1	0.9	23.3	—	0.003	201
2	R	1 650	11	12.4	0.4	16.5	1.3	45.7	0.0	0.8	22.9	—	0.003	194
3	R	1 650	12	17.3	0.3	15.2	1.3	43.0	0.0	0.8	22.1	—	0.003	185
4	R	1 650	12	11.0	0.5	17.7	4.0	47.8	0.1	1.0	18.0	8.51	0.009	193
5	R	1 650	12	11.3	0.6	16.4	3.9	45.0	0.1	1.1	21.6	11.14	0.009	189
6	R	1 650	12	11.8	0.4	16.2	3.7	46.2	0.1	1.1	20.6	10.04	0.009	178
7	R	1 650	12	9.6	0.4	18.0	3.8	49.4	0.0	1.1	17.7	—	0.009	186
8	T	1 550	12	10.6	0.5	17.8	3.8	49.9	0.0	1.2	16.2	9.70	0.004	378
9	T	1 600	12	12.3	0.5	19.2	4.6	48.7	0.0	1.0	13.8	8.40	0.011	187
10	T	1 700	12	11.1	0.4	16.2	3.9	44.7	0.0	1.1	22.5	13.09	0.013	137
11	B2	1 650	10	12.2	0.5	25.4	3.5	46.1	0.0	1.1	11.1	—	0.037	42
12	B2	1 650	11	10.7	0.5	24.6	3.7	48.2	0.0	1.0	11.4	—	0.029	56
13	B2	1 650	12	12.4	0.4	25.1	3.7	46.6	0.0	1.0	10.9	—	0.037	43
14	B2	1 650	12	14.2	0.4	24.1	3.4	45.5	0.0	1.1	11.4	—	0.022	67
15	B2	1 650	12	12.9	0.2	21.8	4.0	46.3	0.0	0.9	13.9	—	0.023	76
16	B2	1 650	12	13.4	0.3	17.2	2.7	34.3	0.0	0.9	31.2	—	0.015	79
17	B2	1 650	12	10.9	0.3	18.2	2.8	36.8	0.0	0.9	30.1	—	0.017	73
18	B4	1 650	12	10.2	0.3	13.5	3.4	55.5	0.0	1.1	16.0	—	0.004	374
19	B4	1 650	12	10.2	0.5	13.2	3.8	51.6	0.0	1.2	19.6	—	0.004	411
20	B4	1 650	12	7.5	0.1	15.1	2.9	67.0	0.0	0.9	6.5	—	0.007	178
21	B4	1 650	12	7.3	0.1	15.4	2.8	66.5	0.0	0.7	7.1	—	0.007	176
22	B4	1 650	12	9.1	0.1	17.0	2.6	65.1	0.0	0.8	5.3	—	0.013	89
23	B4	1 650	12	9.4	0.1	15.4	2.8	61.9	0.0	1.2	9.3	—	0.006	201
24	B4	1 650	12	10.1	0.3	14.5	3.3	56.9	0.0	1.0	14.0	—	0.004	356
25	B4	1 650	12	12.9	0.3	13.1	2.7	54.8	0.0	1.1	15.2	—	0.004	298
26	B4	1 650	12	16.0	0.4	11.1	3.3	48.5	0.0	1.2	19.5	—	0.005	288
27	B5	1 650	12	7.4	0.2	13.0	3.5	64.4	0.0	1.2	10.3	—	0.004	381
28	B5	1 650	12	11.5	0.3	10.6	3.7	48.4	0.0	1.2	24.3	—	0.004	418
29	B5	1 650	12	10.3	0.1	12.9	3.5	62.1	0.0	1.2	9.9	—	0.005	303
30	B5	1 650	12	9.5	0.2	12.0	2.9	63.5	0.0	1.1	10.7	—	0.003	426
31	B5	1 650	12	10.6	0.2	12.5	3.0	61.0	0.0	1.1	11.6	—	0.003	438
32	B5	1 650	12	10.0	0.2	11.3	3.0	63.0	0.0	1.2	11.2	—	0.003	442
33	B5	1 650	12	10.8	0.2	12.4	3.0	60.4	0.0	1.1	12.1	—	0.003	435

*total iron oxides reported as FeO

ments were conducted using the R series slag composition (Table 2) and 1 650°C experimental temperature to test the reproducibility (Table 4). The measured L_P was found to be repeatable as shown by the seven repeats conducted at 1 650°C (see Fig. 3). The mean L_P measured at 1 650°C was 190 with a standard deviation of 7 (3.7% of the L_P value).

A number of experimental series were conducted to test the effect of temperature (denoted T, in Table 4) and basicity (denoted B, in Table 4) on the L_P . The measured L_P was found to decrease with increasing temperature (see Fig. 3). The lines on Fig. 3 represent the Assis *et al.* L_P model¹³ predictions using Eq. (9) for the systems studied. The inputs to Eq. (9) were 10 mass% MgO, 3.3 mass% P_2O_5 , 1 mass% MnO, as well as the measured v-ratio and FeO contents. The observed effect of temperature on L_P was consistent with what has been previously reported and what would be expected from predictions using the Assis *et al.* L_P model.¹³ Generally, the temperature effect is explained by consideration of the Gibbs free energy change (Eq. (3)) for the formation of P_2O_5 (Eq. (2)). From Eq. (3), it can be seen that increasing temperature increases the ΔG° term, and results in a smaller equilibrium constant. From an enthalpy perspective, this smaller equilibrium constant can be considered a reaction response of an exothermic reaction to increasing temperature. For all other factors being equal the smaller equilibrium constant would result in less phosphorous being removed from the steel and a lower L_P .

The effect of slag basicity (v-ratio) on the measured L_P for

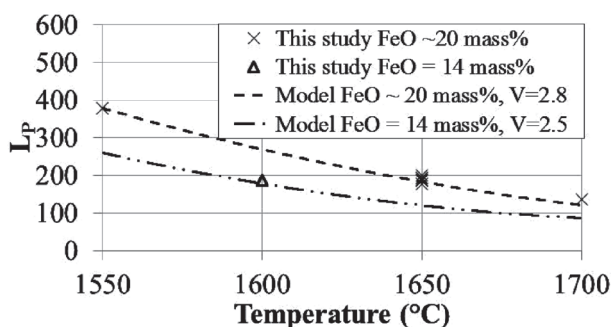


Fig. 3. Measured L_P versus temperature behaviour. The lines represent calculated L_P values using the Assis *et al.* L_P Model.¹³

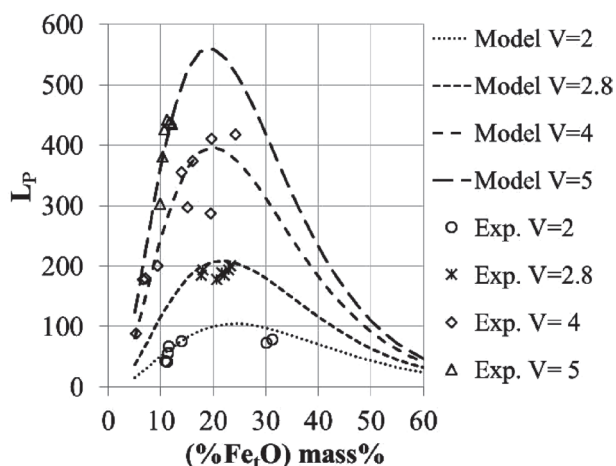


Fig. 4. Measured L_P versus (FeO) at 1 650°C. The lines represent calculated L_P values using Assis *et al.* L_P Model.¹³

a range of FeO concentrations in the slag is given in Fig. 4. Also shown in Fig. 4 are the predicted L_P values for a range of v-ratios and FeO using the Assis *et al.* model¹³ and the inputs stated previously. From this figure, it can be seen there was excellent agreement between the measured L_P and predicted L_P . In both the measurements and, in particular, the model¹³ predictions, a maximum L_P is achieved at approximately 20% FeO for a given v-ratio. This behaviour is consistent with the reported findings of other studies^{25–28} and the maximum represents an optimum slag composition that is informed by the dual requirements of a high oxygen potential and high basicity (v-ratio) on dephosphorisation. This behaviour is best understood by consideration of the reaction given in Eq. (1). From this equation, it can be seen that phosphorous removal from the metal to slag to give a high L_P at a fixed temperature requires both high basicity and high oxygen potential. The oxygen for dephosphorisation in this study is being supplied by the FeO in the slag. Increasing this value beyond approximately 20% FeO causes a drop off in the L_P due to the FeO dilution effect on the basicity of the slag.

While not the goal of this study, the good agreement with the L_P measurements and Assis *et al.* L_P model¹³ predictions shown in Figs. 3 and 4 is encouraging and independently validates the model for use in BOS-type slags. Further, it confirms and extends its applicability to higher temperatures of up to 1 700°C.

3.2. Assessment of the System pO_2 for the $C_{PO_3^{2-}}$

From Eq. (11) it can be seen that the system pO_2 is required for calculating the $C_{PO_3^{2-}}$. In this study, the primary approach used was to calculate the pO_2 from MTDATA²⁹ for the slag-metal systems studied. MTDATA is a thermodynamic software package that uses Gibbs energy minimisation to calculate complex multi-component phase equilibria in gas-liquid-solid systems. The databases used were sgsol, nplox4_nis and htlox5_mnp, with nplox4_nis and htlox5_mnp custom databases prepared for the University of Wollongong. To validate the MTDATA calculations, selected slags were analysed for Fe^{2+} and Fe^{3+} to establish the pO_2 and compared to the MTDATA values (see Fig. 5). The Fe^{2+}/Fe^{3+} ratio is also affected by slag composition/basicity. In the experimental program the slag composition was allowed to change/reach equilibrium. Further, an $\gamma_{(Fe_2O_3)}$ measured in a representative slag system (*i.e.* CaO–SiO₂–MgO– P_2O_5 –FeO–Fe₂O₃) was used in the analysis. The use of this $\gamma_{(Fe_2O_3)}$ will to a large degree mitigate any slag composition/basicity effects. From Fig. 5, it can be seen that the measured values were in good agreement with those predicted and that MTDATA is suitable for predicting pO_2 in the metal-slag systems studied.

The pO_2 was calculated from the slag analysis by consideration of the iron oxidation reaction given in Eq. (15).



The equilibrium constant K for the reaction can be obtained from Eq. (16)

$$K = \frac{\gamma_{(Fe_2O_3)} N_{(Fe_2O_3)}}{a_{Fe}^2 \cdot p_{O_2}^{3/2}} \quad (16)$$

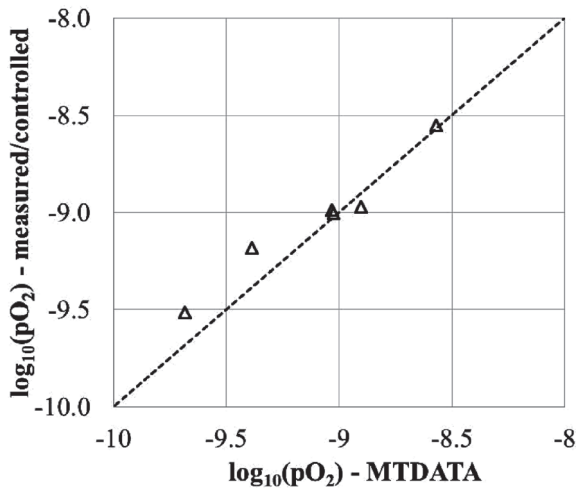


Fig. 5. Comparison of the pO_2 obtained from the Fe^{2+} and Fe^{3+} slag analysis with that of the pO_2 calculated by MTDATA. The dotted line represents a 1 to 1 correspondence.

where $\gamma_{(Fe_2O_3)}$ is the activity coefficient of Fe_2O_3 , $N_{(Fe_2O_3)}$ is the mole fraction of Fe_2O_3 in slag and a_{Fe} is the activity of liquid Fe.

Assuming a_{Fe} has a value of 1 and rearranging for pO_2

$$pO_2 = \left(\frac{\gamma_{(Fe_2O_3)} N_{(Fe_2O_3)}}{K} \right)^{\frac{2}{3}} \dots \dots \dots (17)$$

$\gamma_{(Fe_2O_3)}$ was obtained from the Basu and Lahiri³⁰⁾ relation given in Eq. (18).

$$\log_{10}(\gamma_{(Fe_2O_3)}) = \frac{7272}{T} + 1.69N_{(FeO)} - 34.18N_{(Fe_2O_3)} - 5.86 \dots \dots \dots (18)$$

Using the Fe^{2+} and Fe^{3+} analysis, and assuming the ions represent FeO and Fe_2O_3 in the slag, (FeO) and (Fe_2O_3) can be calculated from (Fe) from the XRF analysis.

The values for K for the reaction were obtained from the Gibbs free energy of reaction, given in Eq. (19).³¹⁾

$$\Delta G^\circ = -815\,000 + 251.1T \quad J \cdot mol^{-1} \dots \dots \dots (19)$$

3.3. Calculation of the $C_{PO_4^{3-}}$

The $C_{PO_4^{3-}}$ was calculated from Eq. (14) using the experimentally measured L_P and the pO_2 determined from MTDATA, where $K_{[P]}$ was determined using Eq. (13) and ideal Henrian behaviour was assumed for [P]. The $C_{PO_4^{3-}}$ was presented as $\log_{10}(C_{PO_4^{3-}})$ versus temperature, v-ratio and Λ in Figs. 6, 7 and 8 respectively.

In Fig. 6 it can be seen the $C_{PO_4^{3-}}$ of the slag decreases with increasing temperature. This is generally explained in terms of consideration of the stability of the PO_4^{3-} ion in the slag, similar to that given for the effect of temperature on the L_P earlier.

In Figs. 7 and 8 it can be seen that increasing basicity as expressed as v-ratio or Λ , increases the $C_{PO_4^{3-}}$. The primary difference in the trends shown in these figures is that the $\log_{10}(C_{PO_4^{3-}})$ trend with v-ratio is not linear and is starting to plateau at v-ratios >5 and that when expressed as $\log_{10}(C_{PO_4^{3-}})$ versus Λ the trend is linear. A regression line representing the linearity of the trend is also given in

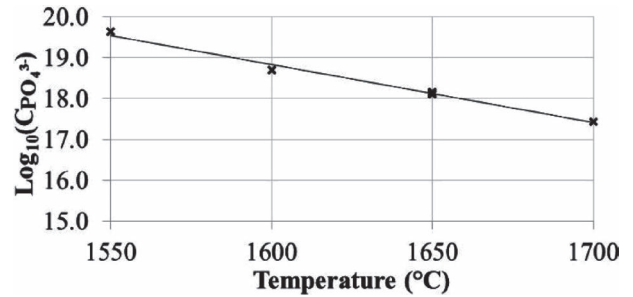


Fig. 6. A plot of $\log_{10}(C_{PO_4^{3-}})$ versus temperature.

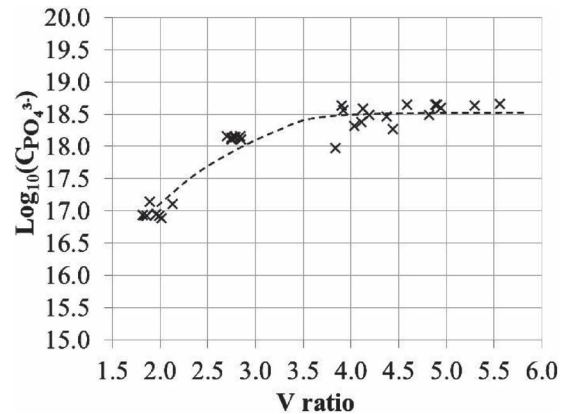


Fig. 7. A plot of $\log_{10}(C_{PO_4^{3-}})$ versus v-ratio at 1 650°C.

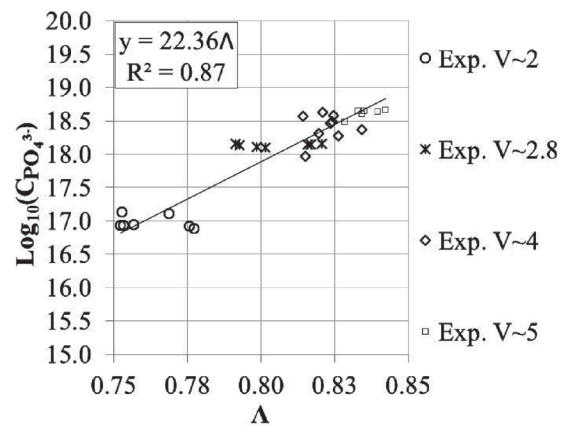


Fig. 8. A plot of $\log_{10}(C_{PO_4^{3-}})$ versus Λ at 1 650°C.

Fig. 8. The deviations from the fitted lines in both Figs. 7 and 8 are thought to represent the inherent uncertainty in the experimental method and the thermodynamic data. The choice of v-ratio and Λ to represent slag basicity is purely for convenience. The increase of $C_{PO_4^{3-}}$ with increasing basicity shown in Figs. 7 and 8 is consistent with what other researchers have reported^{11,14-16,18,32-46)} and can be explained by consideration of Eq. (10) where it can be seen increasing the basicity, represented in this equation as the O^{2-} ion, moves the reaction to the right, i.e. stabilises the PO_4^{3-} ion.

There are some concerns expressed in the literature about the possibility of a phosphorous dimer ($P_2O_7^{4-}$) being formed^{32,41,42,47)} in the slag as opposed to the (PO_4^{3-}) monomer. This has a greater probability to occur at higher

(P) concentrations and lower basicity. In this study the (P) levels were <3 mass% (*i.e.* <6.87 mass% P₂O₅). At these levels the (PO₄³⁻) monomer is the predominant phosphorus oxide structure.^{32,41,42,47)}

3.4. Industrial Relevance

Improved understanding of C_{PO₄³⁻} allows steelmakers to optimise hot metal and steel processing to minimise flux usage and minimise re-processing of out of specification heats. For example, in the BOS, steelmakers often use the predicted slag composition from a heat and mass balance of all charged materials (hot metal, scrap, fluxes, iron ore *etc.*) to determine the C_{PO₄³⁻} of a slag. If this capacity is insufficient to achieve the required grade [P] specification, further flux additions are added until the predicted [P] meets the steelmaking requirements.

Many steelmakers have developed their own C_{PO₄³⁻} (or equivalent) models using various basicity expressions, including v-ratio and Λ . The relationships between C_{PO₄³⁻} and v-ratio (Fig. 7) or Λ (Fig. 8) are both useful for steelmakers. While use of the v-ratio is appealing due to its simplicity and ubiquitous availability, the non-linear behaviour of log₁₀(C_{PO₄³⁻}) with v-ratio may cause issues in application. The less complex, simple linear trend of log₁₀(C_{PO₄³⁻}) with optical basicity may be more readily applied. The use of Λ as opposed to v-ratio also offers the added benefit of accounting for the slag basicity effects of all the oxides, and not just CaO and SiO₂.

4. Conclusions

The L_P was measured for slags in CaO–SiO₂–MgO–Fe₂O₃–(MnO–Al₂O₃–TiO₂–P₂O₅) over temperatures representative of BOS. The measured L_P values were in good agreement with the recently published L_P model proposed by Assis *et al.*¹³⁾ The good agreement between the model and the measured data can be considered validation of the model for use at the higher temperatures and slag basicities used in industrial processes.

The pO₂ for the experiments was also evaluated for the slag-metal systems studied. This was carried out from both a thermodynamic modelling, using MTDATA, and measurement approach. Agreement from both approaches was excellent and allowed evaluation of C_{PO₄³⁻} of the slag from the measured L_P.

Simple correlations of C_{PO₄³⁻} against common slag basicity measures (v-ratio and Λ) were found.

C_{PO₄³⁻} increased with increasing basicity. The increase was not linear with v-ratio but was linear with Λ .

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